SignVaani: A Landmark-Based Real-Time Indian Sign Language to Multilingual Speech Translation System with Signer-Independent Evaluation

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DRAFT TEMPLATE: all values marked [XX] must be replaced with your own measured results before submission.

***Abstract*—Indian Sign Language (ISL) is used by millions of deaf and hard-of-hearing people in India, yet automatic recognition tools overwhelmingly target American Sign Language, and public ISL data remains scarce. This paper presents SignVaani, a real-time system that recognises isolated ISL signs from a commodity webcam, assembles them into grammatical sentences, and produces speech in English and Indian languages. Instead of processing raw RGB video, the system extracts hand, body and selected facial landmarks with MediaPipe Holistic, normalises them relative to the signer's shoulders, and classifies landmark sequences with a lightweight bidirectional GRU with attention pooling. A motion-energy gate combined with an explicit idle class performs sign spotting on a continuous stream, and an ISL-grammar-aware rule engine converts gloss sequences into English before translation with IndicTrans2 and synthesis with Indic-TTS. We also contribute a domain-specific ISL keypoint dataset of [XX] hospital-communication signs recorded from [XX] signers. Under a leave-one-signer-out protocol the system achieves [XX]% top-1 accuracy, compared with [XX]% under a signer-dependent split, and runs at [XX] FPS with [XX] ms end-to-end latency on a laptop CPU without internet access.**

***Index Terms*—Indian Sign Language, sign language recognition, MediaPipe, pose landmarks, GRU, assistive technology, text-to-speech, low-resource languages.**

I. INTRODUCTION

Sign languages are complete natural languages with their own vocabulary and grammar. In India, the Census of 2011 recorded over five million people with hearing disabilities, while the number of certified ISL interpreters is only in the hundreds. As a result, deaf individuals routinely face communication barriers in hospitals, banks, railway stations and government offices, where hearing staff do not understand ISL.

Automatic sign language recognition (SLR) can reduce this gap. However, most datasets and tools address American Sign Language (ASL) [1], [2]. ISL differs from ASL in vocabulary, frequent use of two hands, non-manual markers such as eyebrow movement and head shakes, and a largely subject-object-verb word order. Existing ISL resources such as INCLUDE [3] and iSign [5] are valuable but either lack an end-to-end speech interface or require heavy models that do not run in real time on low-cost hardware.

Furthermore, many student and applied SLR systems evaluate on random splits in which the same signer appears in both training and test sets. This produces inflated accuracy that does not reflect deployment, where the user is always a new person. This paper addresses these gaps with the following contributions:

- A complete, offline, CPU-only pipeline from webcam to multilingual speech for ISL.
- A compact landmark representation combining hands, upper body and selected facial points, with ablations showing the contribution of each component.
- A motion-energy and idle-class based sign spotting method for continuous streams.
- An ISL-grammar-aware gloss-to-sentence module that converts ISL word order into fluent English.
- A new domain-specific ISL keypoint dataset and a signer-independent evaluation protocol.

The rest of the paper is organised as follows. Section II reviews related work, Section III describes the proposed method, Section IV details the experimental setup, Section V presents results, and Section VI concludes.

II. RELATED WORK

A. Sensor- and Vision-Based SLR

Early systems used data gloves and accelerometers, which are accurate but intrusive and ignore facial expression. Vision-based methods progressed from hand-crafted features to deep networks. 3D CNNs such as I3D trained on WLASL [1] learn directly from RGB video but require large datasets and GPUs. Surveys by Koller [6] and Rastgoo et al. [7] note that data scarcity and signer variation remain the main challenges.

B. Pose-Based SLR

Pose-based methods operate on body keypoints estimated by tools such as OpenPose or MediaPipe [8], [9]. SPOTER [10] applies a transformer to pose sequences and performs well on small datasets, and SAM-SLR [11] combines skeleton graphs with other modalities. OpenHands [4] released pose-based pretrained models for several sign languages, including ISL. Pose input is compact, background-invariant and privacy-preserving, which motivates our design.

C. Indian Sign Language Resources

INCLUDE [3] provides 4,287 videos of 263 ISL word signs. ISL-CSLTR [12] offers sentence-level recordings, and iSign [5] is a large ISL-English benchmark built from online videos. Continuous sign language translation [13], [14] remains an open problem, with modest accuracy on ISL. Our work therefore focuses on a reliable, domain-limited vocabulary with an end-to-end speech interface, rather than open-vocabulary translation.

III. PROPOSED METHODOLOGY

Fig. 1 shows the overall pipeline. Every stage runs locally, and only landmark coordinates (not video) are retained.

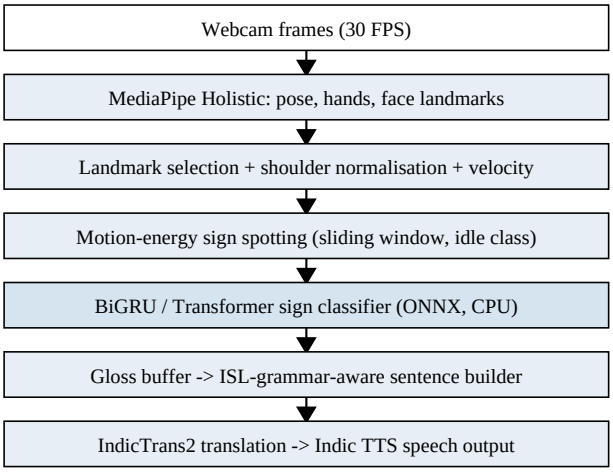

Fig. 1. Overall architecture of the proposed SignVaani system.

A. Vocabulary and Data Collection

We selected [XX] signs for hospital communication (e.g. PAIN, FEVER, DOCTOR, MEDICINE, WATER, TOILET, HELP, WHERE) together with an explicit IDLE class containing resting hands and natural non-sign movements. Signs were verified against the ISLRTC dictionary and by [an ISL interpreter / deaf school partner]. [XX] signers (age [XX]-[XX], [XX] female, [XX] left-handed) each performed every sign [XX] times at 30 FPS under varied lighting and backgrounds. Written consent was obtained, and only keypoints are publicly released.

B. Landmark Extraction and Selection

MediaPipe Holistic outputs 33 pose, 2×21 hand and 468 face landmarks per frame. We retain nine upper-body pose points (nose, eyes, shoulders, elbows, wrists), both hands, and twelve facial points around the lips, eyebrows and eye corners that carry non-manual information, giving $K = 63$ landmarks with (x, y, z) coordinates. Missing hands are linearly interpolated across gaps of up to [XX] frames and otherwise zero-filled with a binary visibility mask.

C. Normalisation and Feature Construction

To achieve invariance to signer position and body size, each frame is centred on the shoulder mid-point c_t and scaled by the mean shoulder width s:

$$p'_{t,k} = (p_{t,k} - c_t) / s\,, \quad v_{t,k} = p'_{t,k} - p'_{t-1,k} \quad (1)$$

Positions and velocities are concatenated into a 378-dimensional vector per frame, and each sequence is resampled to $T = 32$ frames. Training-time augmentation includes rotation (±15°), scaling (0.9-1.1), temporal speed change (0.8-1.2×), Gaussian coordinate noise and random hand drop-out.

D. Sign Classification Model

The classifier (Fig. 2) projects each frame to 256 dimensions with a linear layer and layer normalisation, followed by a two-layer bidirectional GRU (hidden size 128). Temporal attention pooling computes

$$\alpha_t = softmax(w^T h_t), \quad z = \Sigma_t \alpha_t h_t \quad (2)$$

and a final linear layer outputs probabilities over N+1 classes (N signs plus IDLE). The model has about [XX] M parameters. As a comparison we also train a 4-layer Transformer encoder and a temporal 1D-CNN. All models are trained with AdamW (learning rate 10^{-3}, cosine decay), label-smoothed cross-entropy and early stopping, and exported to ONNX for CPU inference.

Fig. 2. Baseline BiGRU classifier with attention pooling.

E. Real-Time Sign Spotting

A rolling buffer stores recent frames. The hand motion energy E_t, defined as the mean wrist speed, opens a candidate window when it exceeds a threshold τ_{on} and closes it after E_t stays below τ_{off} for eight frames. The window is classified, and a sign is accepted only if its probability exceeds 0.8 and the predicted class is not IDLE. A 0.5 s cool-down suppresses duplicate detections.

F. Gloss-to-Sentence Generation

ISL glosses omit articles and tense markers and often follow topic-comment order. A rule engine maps gloss patterns to English templates; for example, [PRONOUN, SYMPTOM, BODYPART] maps "ME PAIN STOMACH" to "I have stomach pain.", and a WH-sign with raised eyebrows produces a question. An optional small local language model polishes grammar but is restricted to the recognised glosses to prevent hallucinated content.

G. Translation and Speech Synthesis

English sentences are translated into Hindi and Telugu with IndicTrans2 [15] or a verified domain phrase table, and spoken using Indic-TTS [16]. Generated audio is cached for frequently used phrases.

IV. EXPERIMENTAL SETUP

Protocols. We report two protocols: (i) *signer-dependent*, a random 70/15/15 clip split, and (ii) *signer-independent*, leave-one-signer-out (LOSO) cross-validation, where the test signer never appears in training. We also evaluate on the overlapping subset of INCLUDE [3] to assess cross-dataset generalisation.

Metrics. Top-1 and top-3 accuracy, macro F1, false triggers per idle minute, end-to-end latency, and a human rating (1-5) of generated sentences. Usability is measured with the System Usability Scale (SUS).

Hardware. Training on [GPU / Google Colab T4]; inference on a laptop with an Intel Core i5-[XX] CPU, 8 GB RAM and a 720p webcam, without GPU or internet.

V. RESULTS AND DISCUSSION

A. Recognition Accuracy

TABLE I
RECOGNITION RESULTS ([XX] SIGNS + IDLE)

Model	Signer-dep. Top-1 (%)	LOSO Top-1 (%)	LOSO Macro-F1
1D-CNN	[XX]	[XX]	[XX]
Transformer	[XX]	[XX]	[XX]
BiGRU + Attn. (ours)	[XX]	[XX]	[XX]
Ensemble	[XX]	[XX]	[XX]

Table I compares models. As expected, accuracy drops from the signer-dependent to the signer-independent setting, by [XX] percentage points for the BiGRU, which highlights why random-split results overstate real-world performance. [Discuss which model performed best and why, e.g. the GRU's advantage on small data.]

B. Ablation Study

TABLE II
ABLATION ON FEATURE COMPONENTS (LOSO, BiGRU)

Configuration	**Top-1 (%)**	Δ
Hands only	[XX]	—
+ Upper-body pose	[XX]	[+XX]
+ Facial points	[XX]	[+XX]
+ Normalisation	[XX]	[+XX]
+ Velocity features	[XX]	[+XX]
+ Augmentation (full model)	[XX]	[+XX]

Table II shows the contribution of each component. [Describe, e.g., that body pose helps signs whose meaning depends on location relative to the body, and facial points help distinguish questions.]

C. Confusion Analysis

The most frequently confused pairs were [SIGN A / SIGN B] and [SIGN C / SIGN D], which share a hand shape and differ mainly in [movement direction / palm orientation]. [Insert the confusion matrix as Fig. 3 and discuss failure cases.]

D. Real-Time Performance

TABLE III
RUNTIME ON LAPTOP CPU (NO GPU)

Stage	**Time / window (ms)**
Landmark extraction (per frame)	[XX]
Classifier (ONNX)	[XX]
Sentence + translation	[XX]
Speech synthesis	[XX]
End-to-end latency	[XX]
Throughput	[XX] FPS

The full pipeline runs at [XX] FPS with [XX] false triggers per idle minute, confirming suitability for real-time use on low-cost hardware.

E. User Study

[XX] deaf participants and [XX] hearing hospital staff used the system in role-played scenarios. Generated sentences received a mean rating of [XX]/5 and the SUS score was [XX]. Participants valued [feedback] and requested [feedback].

F. Limitations

The system recognises a limited vocabulary of isolated signs and does not handle full ISL grammar such as spatial referencing and classifiers. Regional sign variants and fast fingerspelling remain challenging, and the dataset, although signer-diverse, is small. The tool is intended to support, not replace, human interpreters.

VI. CONCLUSION AND FUTURE WORK

We presented SignVaani, an offline, real-time ISL-to-speech system built on compact hand, body and face landmarks. Signer-independent evaluation, ablations and a user study show that a lightweight recurrent model with careful normalisation and an explicit idle class can deliver practical accuracy on a laptop CPU. Future work includes continuous sign language translation trained on iSign, a reverse speech-to-sign avatar, an Android deployment and user-specific personalisation through on-device fine-tuning.

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